

Tide retrospective: *the large- t asymptote of the first-cumulant FDT*

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Abstract

The companion to the cross-susceptibility asymptote, completing both $t \rightarrow \infty$ limits the FDT-capstone tide deferred: the first-cumulant fluctuation–dissipation limit

$$\left. \frac{\partial}{\partial h} \right|_0 \langle x \rangle_h = -t \operatorname{Var}_0(x) \longrightarrow -\frac{1}{\lambda}.$$

The FDT identity is phrased with the abstract `Threepoint` covariance while the asymptotic input uses the concrete `Laplace` covariance; the tide also lands the small definitional bridge between them. New file `AnharmonicFDTMeanAsymptotic.lean`; builds clean first try, no `sorry/axiom/native_decide`.

1 Setting

The cross-susceptibility asymptote tide¹ closed one of the two limits deferred by the FDT capstone, the one that composed cleanly because both factors used `Threepoint.kappa3`. The *mean* limit did not: the FDT identity gives $\partial_h \langle x \rangle_h|_0 = -t \cdot \text{Threepoint.gibbsCov} \cdots 0$, but the available asymptotic, `cov_self_anharmonic_asymptotic`, is stated for `Laplace.gibbsCov`. This tide supplies the missing definitional bridge and then closes the limit.

2 The thing formalised

*For the anharmonic potential, $\operatorname{deriv}(h \mapsto \langle x \rangle_h)(0) \rightarrow -1/\lambda$ as $t \rightarrow \infty$; and, as reusable infrastructure, the abstract *Threepoint* Gibbs expectation/covariance at $h = 0$ (with $A = \operatorname{id}$, against Lebesgue) equals the concrete *Laplace* one.*

3 Proof strategy

Bridge. `Threepoint.gibbsExp` at $h = 0$ unfolds to $(\int \varphi e^{-t(L+0 \cdot x)})/(\int e^{-t(L+0 \cdot x)})$ and `Laplace.gibbsExpectation` to $(\int \varphi e^{-tL})/\text{partitionFunction}$; `simp [zero_mul, add_zero]` collapses the $0 \cdot x$ and the two coincide. The covariance bridge is then three rewrites of the expectation bridge. *Limit.* As in the cross-susceptibility tide: `cov_self_anharmonic_asymptotic` gives $t \cdot \operatorname{Var} \rightarrow 1/\lambda$, so its negation $-(t \cdot \operatorname{Var}) \rightarrow -1/\lambda$; on $t > 0$ the FDT identity (after the covariance bridge) rewrites $\operatorname{deriv}(\langle x \rangle_h)(0)$ to $-t \cdot \text{Laplace.gibbsCov}$, and `Tendsto.congr'` transfers the limit.

¹2026-05-31-tide-anharmonic-fdt-asymptotic.

4 Roadblocks and resolutions

None — first-try build. The only subtlety was anticipated: the FDT identity and the asymptotic live in different covariance namespaces, so the bridge had to be proved first. It is a pure `simp [zero_mul, add_zero]` after unfolding, because the two definitions agree once the perturbation term vanishes at $h = 0$.

5 What was Mathlib, what was new

Mathlib. The `Tendsto` combinators and `atTop` helper², plus `HasDerivAt.deriv`, `simp`, `ring`.

Seabed (reused). The unconditional first-cumulant FDT identity and the covariance asymptotic.³

New. The `Threepoint-to-Laplace` expectation/covariance bridge (reusable) and the mean asymptote. About 50 lines.

6 Lessons

- **A namespace bridge is often the whole tide.** Two formalisations of “the same” object (here the unperturbed Gibbs covariance) can block a clean composition; the bridge is a short `unfold + simp` once the defining expressions are made to coincide.
- **The bridge is reusable.** Any future result that needs the abstract `Threepoint` machinery’s $h = 0$ specialisation to meet the concrete `Laplace` asymptotics can now cite it.

7 Follow-ups

- **Promote `amgm_t_abs_x` to `lean-common`** — the last remaining `laplace-arc` follow-up.
- **A fresh seabed.** The anharmonic 1D FDT / cumulant programme is now complete (both asymptotes, all cumulants through κ_4 in three bases); the next substantial step is a different research area.

²`Filter.Tendsto.neg`, `Filter.Tendsto.congr'`, `eventually_gt_atTop`.

³`cov_self_anharmonic_asymptotic`.